SHRUTI SHARMA

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SUMMARY

AI/ML-focused engineering student who builds machine learning, NLP and generative AI systems end to end, from research and prototyping to deployment. Hands-on with LLM integration, information retrieval (embeddings, FAISS/HNSW, BM25/TF-IDF), recommender systems and ML pipelines served through REST APIs. 2nd place at HACK60 (IIT Mandi x HCLTech) for an LLM-powered, real-time personalization engine with sub-500ms latency.

TECHNICAL SKILLS

Languages: Python, SQL, C++, JavaScript, Java (familiar), HTML5, CSS3
ML & NLP: Machine Learning, scikit-learn, NLP, Reinforcement Learning, recommender systems, model evaluation (Recall@K, MRR, NDCG), OpenCV
Generative AI & IR: LLM integration (Groq / Llama 3, Ollama), RAG, LLM re-ranking, sentence embeddings (SBERT), FAISS/HNSW vector search, TF-IDF, BM25, cosine similarity
Engineering & Deployment: REST API design, Flask, SQL/SQLite, Git/GitHub, GitHub Actions (CI/CD), automated testing, Streamlit, Plotly, Vercel, Streamlit Cloud
Core CS: Data Structures & Algorithms, OOP, Software Design, Computer Organisation

PROJECTS

NewsLens – Hyper-Personalization Engine *2nd Place, HACK60 (IIT Mandi x HCLTech)*
Python, LLMs (Groq / Llama 3), Reinforcement Learning, TF-IDF, OpenCV
- Built a working AI prototype: a hybrid recommender combining Thompson Sampling RL, TF-IDF and collaborative filtering to personalize news feeds.
- Integrated an LLM (Groq / Llama 3) as a real-time re-ranking module, with OpenCV gaze tracking feeding live mood signals into content ordering.
- Trained and validated on 156K interactions from the MIND dataset; optimized inference to sub-500ms latency, 4x faster than the target benchmark.

Semantic Retrieval Engine *Embeddings + Vector Search*
Python, sentence embeddings (SBERT), FAISS / HNSW, REST API
- Developed a semantic search engine that retrieves documents by meaning using transformer embeddings and an approximate-nearest-neighbour index, served through a REST API.
- Evaluated dense retrieval against TF-IDF and BM25 baselines using Recall@K, MRR, NDCG and query latency to guide model choice.

FraudShield – Explainable Fraud Detection *GitHub*
Machine Learning, Explainable AI, React, Recharts, Jupyter
- Built an explainable AI-powered credit card fraud detection dashboard with a backend service, an interactive React frontend and notebooks for model development and analysis.
- Designed the dashboard to surface why a transaction is flagged, so analysts can interpret model decisions instead of relying on a bare fraud score.

PaperPilot – Research Paper Assistant *GitHub*
Python, Streamlit, PyMuPDF, SBERT (all-MiniLM-L6-v2), FAISS, Ollama (qwen2.5:3b)
- Built a RAG-based assistant that lets users upload research PDFs and ask natural-language questions, with answers grounded in retrieved passages and page citations (e.g. [Page 4]).
- Implemented semantic search with SBERT embeddings and a FAISS index, and local LLM inference through Ollama so papers stay on the user's machine.
- Added a paper library with per-paper chat history, cross-paper questions, a summary toolkit (TL;DR, methodology, key findings, contributions, limitations) and Markdown export, in a custom dark-themed Streamlit UI.

EDUCATION

Indian Institute of Technology (IIT) Mandi – B.Tech in Data Science Engineering *2024 – Present*
Wisdom Public School, Aligarh (CBSE): Class XII – 94.2%, Class X – 90.5%

ACHIEVEMENTS

- 2nd Place, HACK60 – hackathon organized by IIT Mandi in collaboration with HCLTech.